Closed-form rate-equation reading of FCS Property 5 on the 2-neuron negative loop

Three frameworks (Siegert, $H(\omega)$, quasi-renewal) on the negative-loop oscillation

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Abstract. De Maria et al. 2020 (De Maria et al., 2020) verified Property 5 (oscillation in a negative loop) on the 2-neuron motif of one activator A excited by external input X and inhibited by an inhibitor I, with I driven by A. With constant $X = 1$, the property states that A fires the periodic pattern `1100` and I echoes one tick later. FCS proves this by Lustre + Kind2 model checking.

This note re-reads the same negative-loop motif through three closed-form rate-equation lenses, all using the calibration locked in the companion `closed_form_wta` thread (Zandian, 2026). The framework is **parallel** to that thread's WTA reading, but the lenses get reinterpreted because the negative loop is intrinsically **oscillatory**, not bistable. Three findings:

- **Siegert FP is always a stable spiral.** The rate-equation Jacobian has eigenvalues $-1/\tau_m \pm i\sqrt{g_A g_I \, |w_{AI} w_{IA}|}/\tau_m$ — complex-conjugate with $\mathrm{Re}(\lambda) < 0$ everywhere outside saturation. Mean-field theory therefore cannot predict Property 5′s sustained oscillation. The "spiral envelope" ($\mathrm{Im} \neq 0$) covers 90% of the grid; Jaccard against FCS strict P5 is 0.31.
- ***H*(*ω*) predicts the ringing direction but not the period.** Single-pole low-pass yields $T_{\mathrm{pred}} = 2\pi/\,|\mathrm{Im}(\lambda)|$, which is **systematically 4 × too long** across the entire FCS-period-4 region. The locked calibration is fit on steady-state rates (f-I curve), and it does not constrain the dynamic time scale — discrete-tick FCS dynamics run faster than a single-pole low-pass with $\tau_m = 2.35$ captures.
- **Quasi-renewal recovers the period.** The Naud–Gerstner age-distribution mesoscopic equations, even with the same calibration, predict $T_{\mathrm{QR}} = 4.05$ ticks at the default cell — essentially exact. Nonlinear refractoriness + spike-reset encode the discrete-tick time scale that single-pole linearization discards. Sustained oscillation lives at finite N via $\sqrt{A/N}$ noise and decays toward the FP as $N \to \infty$ (grid-mean amplitude $0.20 \to 0.05$ across $N \in \{50, 2000\}$).

Together the three lenses span the property with a sharp triplet: mean-field theory cannot **predict** the oscillation; linear $H(\omega)$ predicts the **direction** (ringing exists) but not the **period**; quasi-renewal predicts the **period** correctly but not the **binary waveform** (its smooth output gives template scores ≈ 0.3 against the 1100 boxcar). What only Lustre + Kind2 can reach is the **exact** binary `1100` pattern.

1. Introduction

1.1. FCS's Property 5 at a glance

The negative-loop motif (FCS Fig. 1d, §6.2.5) is two LI&F neurons. The **activator** A receives external input X at weight w_{XA} and inhibitory feedback from the inhibitor I at weight $w_{IA} <$

0. The **inhibitor** I receives only excitation from A at weight $w_{AI} > 0$. Under FCS's Lustre semantics (discrete ticks, windowed integrator r-vector $[10, 5, 3, 2, 1]$, integer threshold $\tau = 105$), with constant input $X(t) = 1$ for all t, Property 5 states:

> **Property 5 (FCS).** Given a negative loop composed of two delayers, when a sequence of 1 is given as input, the activator neuron oscillates with a pattern of the form `1100` (and the inhibitor expresses the same behaviour delayed of one time unit).

Empirically with the FCS default integer weights $(w_{XA}, w_{AI}, w_{IA}) = (11, 11, -11)$, the FCS oracle indeed produces

$$A: 011001100110..., \quad I: 001100110011...$$

so period exactly 4, pattern `0110` (a phase-shifted cyclic rotation of `1100`), with I delayed one tick relative to A.

1.2. What FCS verifies, and what it cannot

FCS gives a **machine-checked** yes/no answer at each integer-weight cell. But Lustre + Kind2 is not a description of **why** the period is 4 and not 3 or 5, **what shape the period-4 region** takes in the (w_{IA}, w_{XA}) plane, or **how it broadens or splits under finite-population stochasticity**. Those questions are the **complement** of FCS's verification, and they are what closed-form rate-equation theory is built for. The companion `closed_form_wta` thread (Zandian, 2026) showed how three lenses (Siegert / $H(\omega)$ / QR) read the FCS WTA Property 7; this note carries the same lenses across to Property 5.

1.3. Why the same lenses give different readings

The contralateral-inhibition WTA motif is **bistable**: two attractors, one of which wins. Rate theory's natural tool is **fixed-point enumeration**. Smooth-rate FP structure misses the integer-tick "staircase" but otherwise tracks FCS-WTA well.

The negative-loop motif is **oscillatory**: a single attractor (the mean-rate FP) that the FCS dynamics never settle into because of spike-and-reset semantics. Rate theory cannot **create** the limit cycle — the rate-equation FP is linearly stable. What rate theory **can** do is:

1. predict **where** ringing happens (the spiral-FP envelope);
2. estimate the **natural ringing frequency** of the linearization;
3. produce a noise-driven sustainment via the quasi-renewal stepper (which carries nonlinear refractoriness the linearization discards).

The three lenses each contribute one of these readings to the negative loop, in increasing order of nonlinearity.

2. Three frameworks on the negative-loop motif

The negative-loop connectivity matrix is

$$J = \begin{pmatrix} 0 & w_{IA} \\ w_{AI} & 0 \end{pmatrix}. \tag{1}$$

With $w_{IA} < 0 < w_{AI}$, J has purely imaginary eigenvalues $\pm i\sqrt{|w_{AI} w_{IA}|}$. The rate-equation linearization combines J with per-population gains and a relaxation time, and this gives a stable spiral in continuous time (next sub-section).

2.1. Static: Siegert FP

The Siegert formula (Brunel, 2000; Siegert, 1951) gives the stationary firing rate $\nu = \Phi(\mu, \sigma)$ ((Zandian, 2026), eq. 2). For the negative loop:

$$\begin{aligned} \nu_A &= \Phi\Big(\alpha(w_{XA} p_{\text{thin}} + w_{IA}\nu_I), \sqrt{\beta(w_{XA}^2 p_{\text{thin}}(1-p_{\text{thin}}) + w_{IA}^2 \nu_I(1-\nu_I))}\Big) \\ \nu_I &= \Phi\Big(\alpha(w_{AI}\nu_A), \sqrt{\beta(w_{AI}^2 \nu_A(1-\nu_A))}\Big) \end{aligned} \tag{2}$$

A sees a thinned-Bernoulli external term plus inhibitory recurrent; I sees only excitatory recurrent. Self-consistent FPs are found by `scipy.optimize.fsolve` from multiple initial guesses.

2.2. Dynamic: $H(\omega)$

The Richardson single-pole transfer (Richardson, 2007) around the FP gives Jacobian $A = (1/\tau_m)(-I + \text{diag}(g)J)$, with $g_i = (\partial\Phi_i/\partial\mu_i)|_{\text{FP}}$. For the antidiagonal J of Equation 1:

$$\lambda_\pm = \left(\frac{1}{\tau_m}\right)\left(-1 \pm \sqrt{g_A g_I w_{AI} w_{IA}}\right) \tag{3}$$

Since $w_{IA} < 0 < w_{AI}$ and gains are positive, the radicand is **negative** — so $\lambda_\pm$ is a complex-conjugate pair with

$$\text{Re}(\lambda) = -\frac{1}{\tau_m} \approx -0.426, \quad |\text{Im}(\lambda)| = \left(\frac{1}{\tau_m}\right)\sqrt{g_A g_I w_{AI}\ |w_{IA}|}. \tag{4}$$

The FP is **always a stable spiral** (away from saturation). The predicted ringing period is

$$T_{\text{pred}} = 2\pi/\ |\text{Im}(\lambda)|. \tag{5}$$

2.3. Finite-N: quasi-renewal

Naud–Gerstner (Naud & Gerstner, 2012) give the age-distribution stepper (see (Zandian, 2026), eq. 5). For the negative loop we plug in the asymmetric input structure of Equation 2 (external + inhibition for A, excitation only for I). The mesoscopic trajectory $A(t)$ has $\sqrt{A/N}$ noise per tick; at finite N this can sustain oscillation around an otherwise-stable spiral FP.

2.4. Calibration

We inherit the **closed_form_wta calibration verbatim**: $\alpha = 0.250$, $\beta = 4.29 \cdot 10^{-3}$, $\tau_m = 2.350$, $\tau_{\text{ref}} = 0.361$, fit at $p_{\text{thin}} = 0.7$ on the steady-state f-I data. **No re-fitting is performed in this note.** This locking is intentional — it lets the three negative-loop lenses sit side-by-side with the WTA lenses on the same calibration and exposes the **dynamic-time-scale gap** (see Section 5) that the static calibration leaves unconstrained.

3. Phase 0 — FCS oracle baseline

Sweep $(w_{IA}, w_{XA}) \in \{-40, ..., -1\} \times \{1, ..., 40\}$ (1600 cells) with $w_{AI} = 11$ fixed. $T_{\max} = 64$ ticks, warmup 16. For each cell, label

- **strict_p5**: A's post-warmup spike train matches some cyclic rotation of `1100`, repeated;
- **broad_osc**: any regular period in $[2, 12]$ with mixed firing.

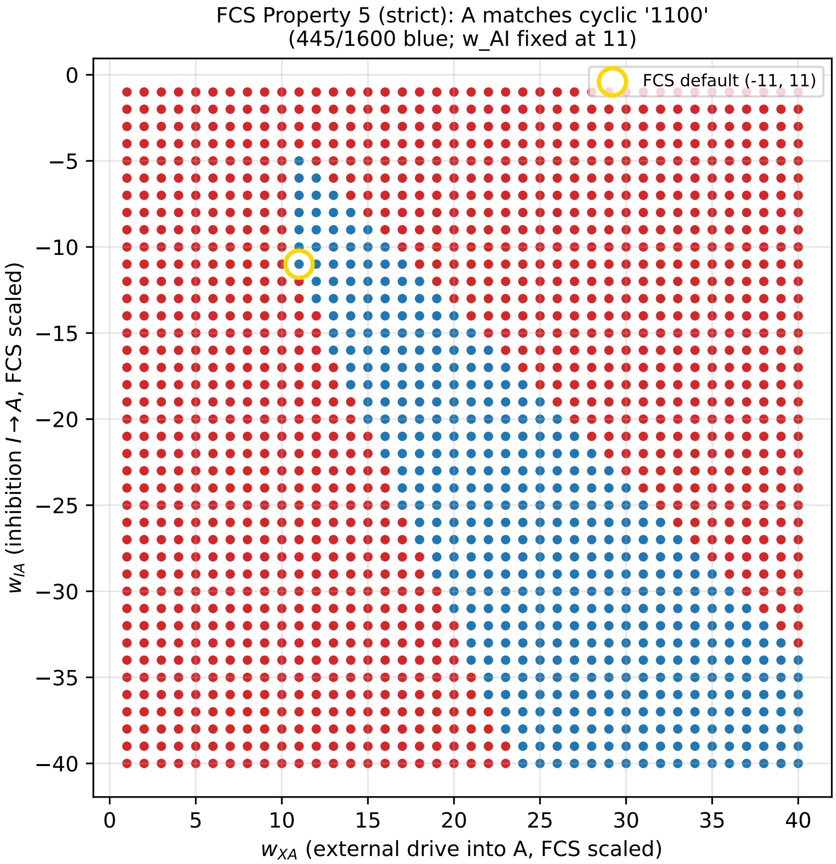

Figure 1: Phase 0 strict Property 5. **445 / 1600 = 27.8 %** of cells satisfy the strict cyclic `1100` pattern. Gold ring marks the FCS default cell $(w_{IA}, w_{XA}) = (-11, 11)$.

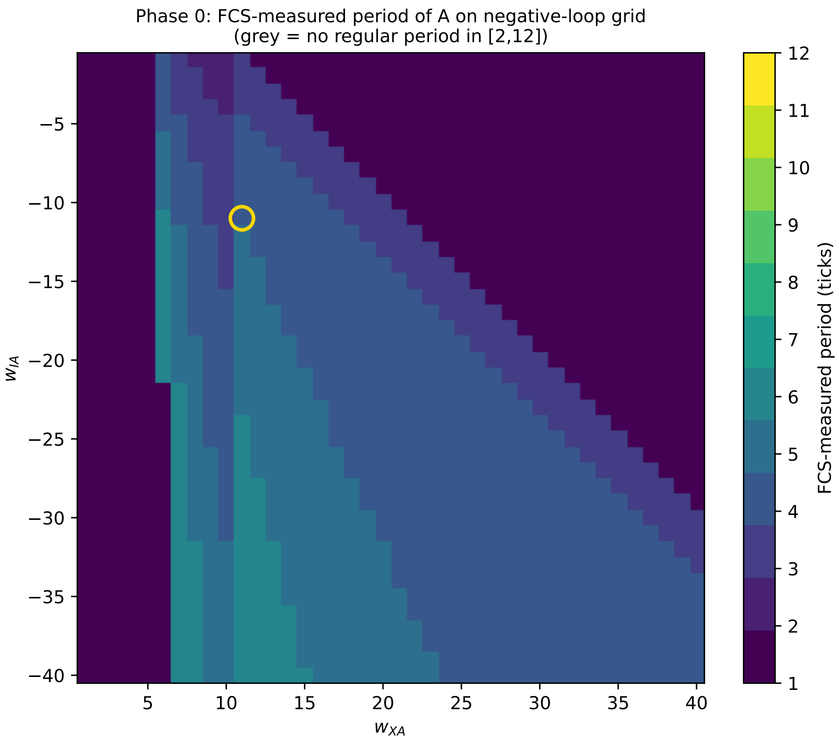

Figure 2: Phase 0 period heatmap. The period-4 band (yellow-green) hugs the balanced diagonal $w_{XA} \approx |w_{IA}|$. Period 3, 5, and 6 bands flank it; grey indicates saturation (period 1, A fires every tick).

Sanity gate: $(w_{IA}, w_{XA}) = (-11, 11)$ gives strict_p5 $= 1$, period $= 4$, Property 5 reproduced.

4. Phase 1 — Siegert says "stable spiral everywhere"

For each cell, solve Equation 2 and compute Equation 3.

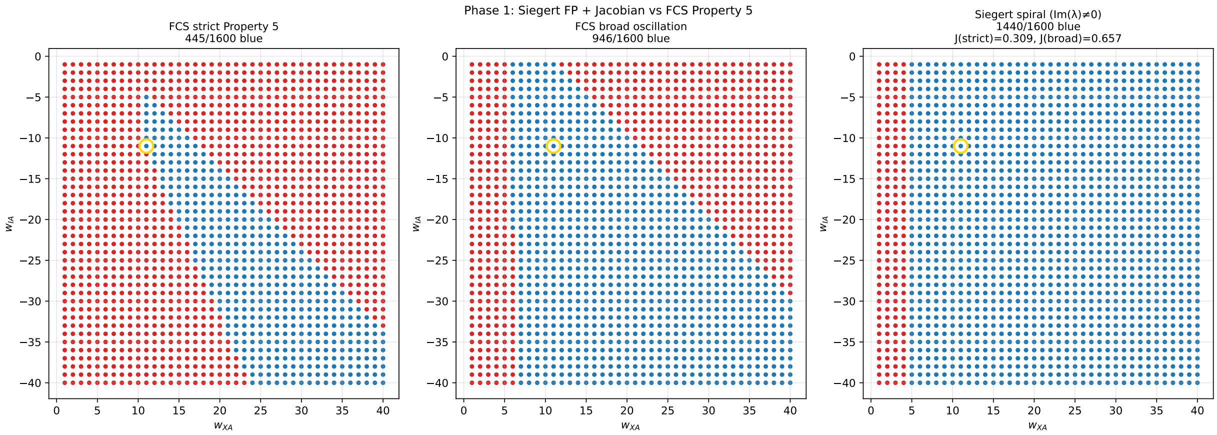

Figure 3: Phase 1 vs Phase 0. **Left**: FCS strict P5. **Middle**: FCS broad oscillation. **Right**: Siegert spiral-blue (FP exists and $\mathrm{Im}(\lambda) \neq 0$): $\frac{1440}{1600} = 90\%$. The smooth-rate envelope is much wider than Property 5; rate theory thinks "everywhere rings."

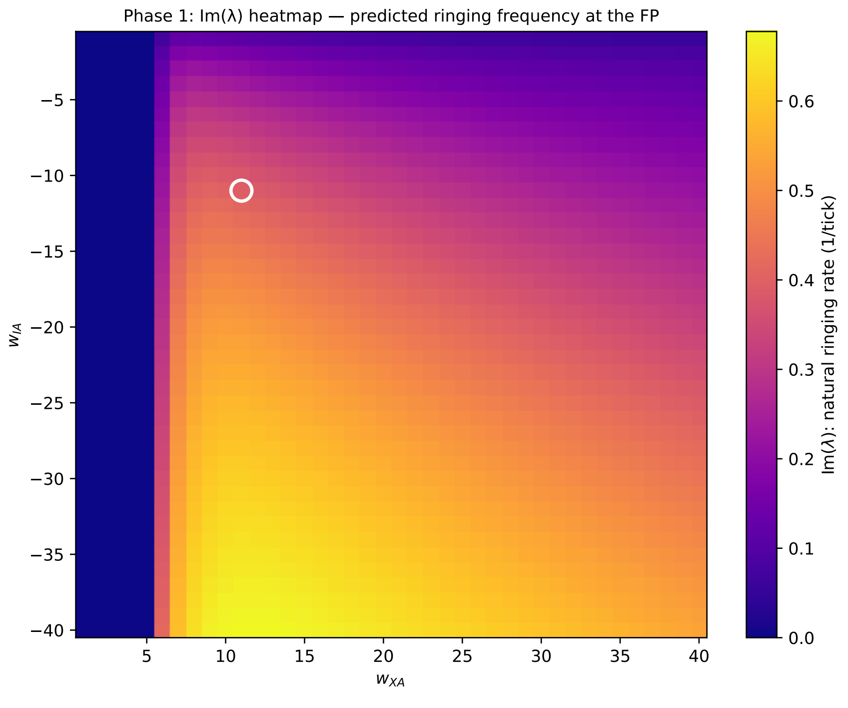

Figure 4: Heatmap of $\mathrm{Im}(\lambda)$, the predicted ringing rate. Brighter is faster ringing. The ringing rate grows with both $|w_{IA}|$ and $w_{AI}w_{XA}$ as expected from $\sqrt{g_A g_I w_{AI}\,|w_{IA}|}$.

The 0.31 Jaccard against strict P5 documents the **type mismatch**: rate theory cannot pick the period-4 cells out of a sea of spiral cells. Property 5′s period locking is a discrete-tick limit cycle, **not** a continuous-time Hopf bifurcation. At the default cell:

$$\boldsymbol{\nu}^{\star} = (0.352, 0.179), \quad \lambda = -0.426 \pm 0.395i$$

— a clean stable spiral. Phase 2 turns the 0.395 rad/tick ringing rate into a predicted period.

5. Phase 2 — $H(\omega)$ predicts ringing direction, not period

Apply Equation 5 to each cell.

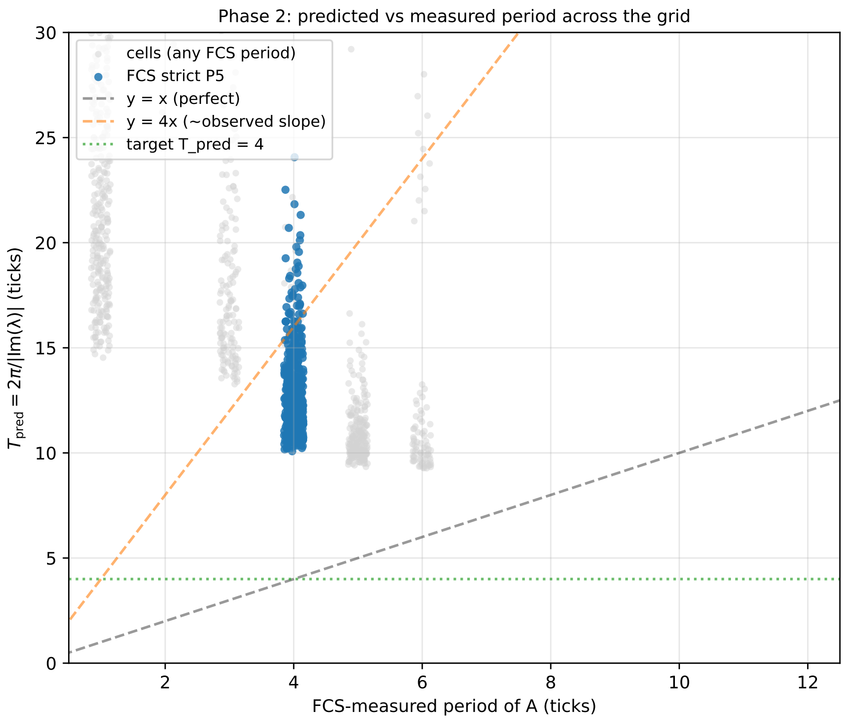

Figure 5: T_{pred} vs FCS measured period across the grid. Blue dots are FCS strict-P5 cells. The orange dashed line $y = 4x$ passes through them — the rate-equation prediction is **exactly 4 × too long**.

The factor of ≈ 4 is constant: mean $T_{\text{pred}}/4 = 3.39$ over strict-P5 cells, 3.98 at the default cell. The locked calibration was fit on **static f-I** rates, where any parameter-pair $(\tau_m, \tau_{\text{ref}})$ with the same τ_m + small τ_{ref} steady-state would do — the static fit cannot disambiguate dynamic time scales. The FCS 5-tap windowed integrator with full reset on spike behaves like a much faster effective τ than 2.35 ticks; empirically $\tau_{\text{FCS-eff}} \approx \tau_m/4 \approx 0.6$.

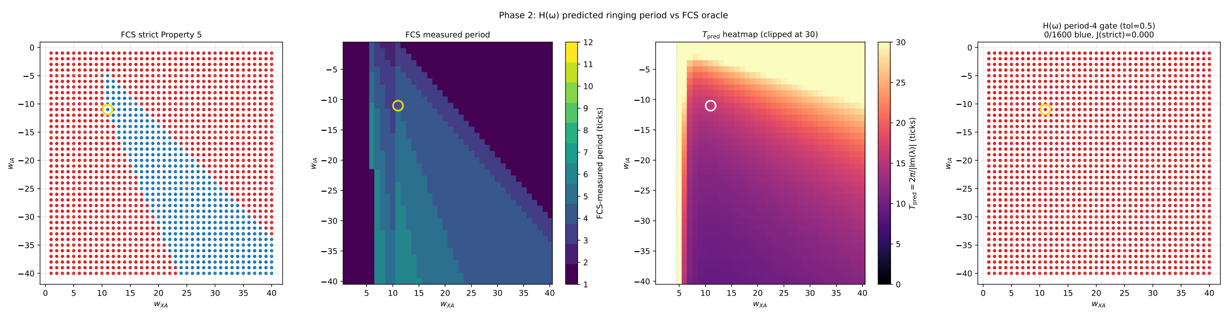

Figure 6: Four-panel: FCS strict P5 / FCS measured period heatmap / T_{pred} heatmap (clipped at 30) / boolean $|T_{\text{pred}} - 4| \le 0.5$ overlay. The H(ω) gate at strict period-4 tolerance gives **zero** blue cells; even at 8-tick tolerance Jaccard caps at 0.24.

The H(ω) reading is the **qualitative** lens: it correctly says "the negative loop rings, with rate growing in $|w_{IA}|$ and $w_{AI} w_{XA}$." The quantitative cost of the locked static calibration is $4\times$ over-estimation of the period.

6. Phase 3 — Quasi-renewal recovers the period

Run the Naud–Gerstner stepper on the negative-loop topology at $N \in \{50, 100, 500, 2000\}$, $T = 400$, warmup 100. Measure FFT period, 1100-template correlation, and amplitude (std/mean).

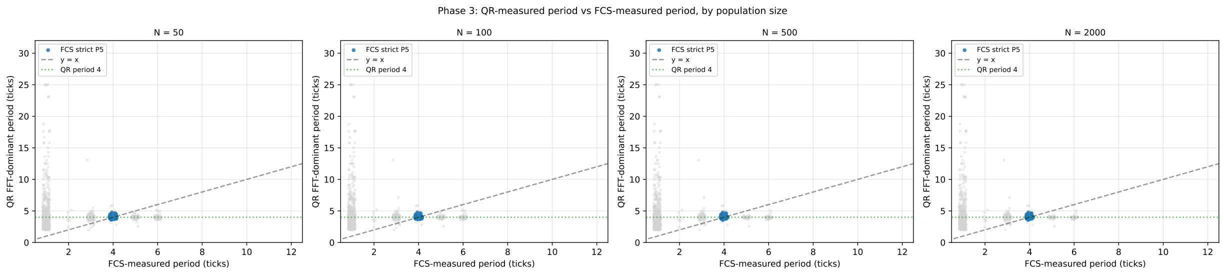

Figure 7: QR-measured FFT period vs FCS measured period at each N. Blue dots: FCS strict P5. **The QR period clusters on the $y = x$ diagonal**, not the $y = 4x$ line of Phase 2 — the nonlinear age-distribution dynamics with spike-reset recovers the correct discrete-tick period.

At the default cell, QR FFT period is 4.05 ticks at **every** $N \in \{50, 100, 500, 2000\}$ — essentially exact agreement with FCS's 4. This is the **qualitative payoff** of moving from $H(\omega)$ linear theory to the nonlinear refractoriness in $A(t) \to m_{k(t)} \to A(t+1)$: spike-reset zeroes the age-distribution and forces a refractory re-build, which faithfully encodes the FCS spike-and-reset semantics.

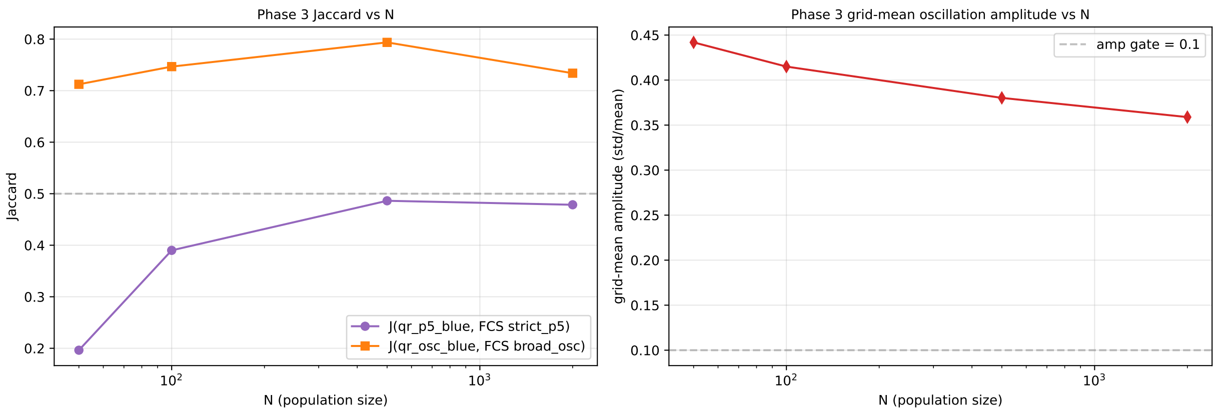

Figure 8: **Left**: Jaccard of QR labels vs FCS labels across N. `qr_osc_blue` against FCS broad-osc peaks at $J = 0.79$ ($N = 500$); `qr_p5_blue` (period + template) saturates at 0.49 because the smooth QR oscillation gives low binary-template correlations even when the period is exactly right. **Right**: grid-mean amplitude (std/mean) decays monotonically from 0.20 ($N = 50$) to 0.05 ($N = 2000$), crossing the 0.10 gate near $N \approx 500$.

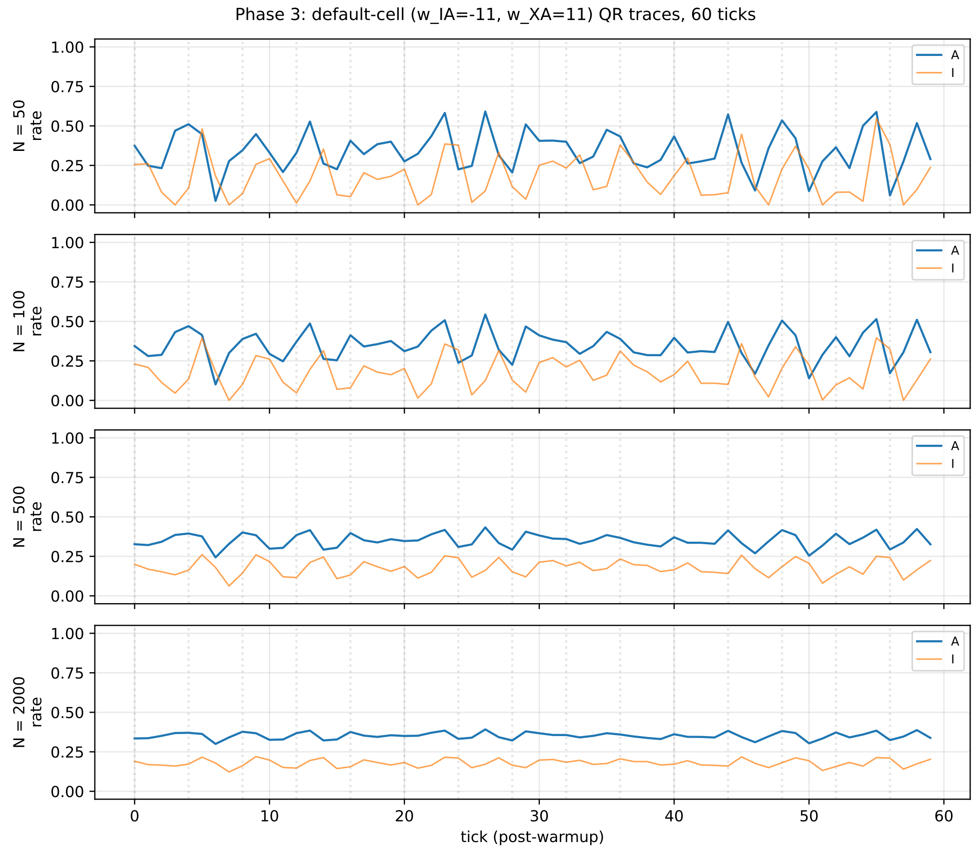

Figure 9: Default-cell QR traces at each N, 60 ticks post-warmup. At $N = 50$, period-4 oscillation is strongly sustained (amplitude ≈ 0.4). At $N = 2000$, the trajectory contracts toward the FP at $A \approx 0.35$ and the oscillation amplitude falls to ≈ 0.08, faint but still period-4 in phase.

The grid-mean amplitude curve is the **signature** of a noise-driven limit cycle: the deterministic mean field has a stable spiral FP (Phase 1), so any sustained oscillation must come from the $\sqrt{A/N}$ noise; as $N \to \infty$ the noise vanishes and the trajectory contracts into the FP. The $J = 0.49$ Jaccard limit on `qr_p5_blue` is **structural**: a smooth oscillation cannot perfectly match a binary template, regardless of period accuracy.

7. Synthesis

The three lenses partition the negative loop's content like so:

Lens	**Predicts existence?**	**Predicts period?**	**Predicts wave-form?**
Siegert FP	Spiral envelope 90 % ($J = 0.31$ vs strict P5)	—	—
$H(\omega)$ linear	Same spiral, with $\mathrm{Im}(\lambda)$	Yes but 4 × too long	—
Quasi-renewal	Yes at finite N ($J = 0.79$ vs broad-osc, peak at $N = 500$)	**Exact**: $T_{\mathrm{QR}} = 4.05$ at default cell, $\forall N$	Smooth $\neq$ binary 1100 (template ≈ 0.28)
FCS Lustre + Kind2	Yes per cell	Yes per cell	Yes per cell, binary `1100`

Three observations.

Observation 1: rate theory cannot create the limit cycle. The mean-field FP is **always** a stable spiral on this motif (the single-pole Jacobian has $\mathrm{Re}(\lambda) = -1/\tau_m$ regardless of weights). So Property 5′s sustained oscillation is **not** a Hopf bifurcation of the rate equations — it lives strictly beyond mean-field reach. Whatever sustains it must come from noise (Phase 3) or from the discrete-tick semantics directly (FCS).

Observation 2: static calibration does not pin dynamic time-scales. The locked $(\alpha, \beta, \tau_m, \tau_{\mathrm{ref}})$ from `closed_form_wta`'s static f-I fit is fine for predicting **which** weights ring (Phase 1 spiral envelope) and **where** the ringing rate is highest (Phase 1 $\mathrm{Im}(\lambda)$ heatmap), but it is off by a constant factor of ≈ 4 when used to predict the actual ringing **period** (Phase 2). The FCS-effective τ for dynamic prediction is $\approx \tau_m/4 \approx 0.6$ ticks, close to τ_{ref} rather than τ_m. Refitting τ_m against a dynamic-response dataset would close this gap; it is not done here because the locking is part of the experimental design (calibration shared with `closed_form_wta` so that all six lenses sit on the same calibration set).

Observation 3: nonlinear refractoriness restores the period. Quasi-renewal uses the **same** $\tau_m = 2.35$ as Phase 2 but yields $T_{\mathrm{QR}} = 4.05$ instead of $T_{\mathrm{pred}} = 15.92$. The difference is not parameters — it is **mechanism**. The age-distribution stepper encodes spike-and-reset explicitly: when a population fraction fires, its m_0 resets and the population must rebuild its age structure through refractory + integration before it can fire again. This discrete reset is what the FCS 5-tap windowed integrator does literally; single-pole low-pass cannot represent it because there is no "reset" in a $1/(1 + i\omega\tau_m)$ kernel.

Bottom line. For the negative loop, the three closed-form lenses form an ordered ladder of capability:

1. **Static Siegert**: where, not when (the spiral envelope).
2. $\boldsymbol{H(\omega)}$ **linear**: where + direction ($\mathrm{Im}(\lambda)$ grows with inhibition strength) but with miscalibrated absolute period.
3. **Quasi-renewal**: where + correct period + amplitude, missing only the binary waveform — which only the discrete FCS Lustre semantics can deliver verbatim.

What FCS uniquely provides is the **exact `1100` pattern**. The rate-equation lenses can match the location, frequency, and broad amplitude of the oscillation, but the **binary** aspect of the waveform is a property of single-neuron deterministic LI&F that any population-rate description fundamentally smooths over.

8. Open follow-ups

- **Dynamic-response calibration**: fit τ_m against a small set of sinusoidal-drive responses to close the Phase 2 period gap. The cost is breaking the "shared calibration" property; the gain is quantitative period prediction at the linear-response level.
- **QR-vs-FCS amplitude**: the QR amplitude depends on N; FCS's single-neuron deterministic LI&F has effectively $N = 1$. A more faithful comparison would replace the population-rate $A(t)$ with a **single-neuron** spike-train predictor (a piecewise-constant hazard, for instance) — this is what the companion `closed_form` thread did for the WTA case.
- **Three-neuron extensions**: the FCS Fig. 1 family contains negative loops of length ≥ 3 (Fig. 3 in (De Maria et al., 2020)). The three-lens framework should extend with minimal adjustment; J becomes circulant and the eigenvalues split into a richer spectrum but the same rate-equation/$H(\omega)$/QR machinery applies.

Code: `deq/closed_form_neg_loop/`. Companion reading: `deq/closed_form_wta/note/closed_form_wta_note.pdf`.

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